

From Candidate Features to a Local Identifiability Result

Continuous feature extraction, sensitivity Jacobian construction, and singular-value analysis at the reference parameter set

Technical writeup, continued

This note continues directly from the companion reconnaissance report (“From a Featureless Global Observable to Mode-Resolved Candidate Inverse Features”), which ended by identifying five candidate modal-response features and naming three explicitly open questions: sub-grid feature precision, parameter sensitivity, and local identifiability. This document answers all three. The colour convention is unchanged: findings in green, open or scope-limited points in red, background definitions in blue.

5 Recap and a terminology correction

The previous report’s reconnaissance identified five candidate features by decomposing the wet EMM response onto the dry circular-plate modal basis: three resonance-like maxima and two modal-coefficient near-zeros,

$$A_{2,0} \text{ maximum, } A_{0,1} \text{ maximum, } A_{1,1} \text{ maximum, } A_{2,1} \text{ near-zero, } A_{0,0} \text{ near-zero,} \quad (1)$$

with discrete grid-level locations of approximately 5.80, 5.65, 6.60, 4.95, and 7.85 rad/s respectively. These were all evaluated at a single trial parameter set,

$$\beta_0 = 4.6985 \times 10^{-5}, \quad \gamma_0 = 1.4548 \times 10^{-3}, \quad R_0 = 0.3830. \quad (2)$$

Terminology correction relative to the previous report. The companion reconnaissance report referred to Equation (2) as the “nominal floe parameters”. That phrase should be read, here and there, as a *reference or test parameter set for inversion-method development*, not as a claim that these values represent a physically typical Antarctic floe. They were chosen because they produce a well-differentiated, easily located response structure across the five features of Equation (1), which is exactly what a reconnaissance stage needs; nothing in that choice was intended to assert physical representativeness, and everything built on this reference point in the present document (sub-grid features, the Jacobian, the identifiability result) should be read as validating the *machinery* at this point, not as a statement about a real floe. Whether this reference point also happens to sit inside a physically plausible range for Antarctic floe geometry and thickness has not been checked and is left open.

6 A continuous feature extractor

The grid-quantisation problem. The previous report’s broad-maximum caveat anticipated exactly the difficulty addressed here: a discrete arg max or arg min, re-searched on a fixed frequency grid at every trial (β, γ, R) , is a step function of the parameters. A small perturbation can leave the extremum on the same grid point (an artificially zero derivative), then cross a threshold and jump by a full grid step. Such a feature is unusable as the observable in a finite-difference Jacobian.

Fixed-anchor sub-grid interpolation. For each of the five features, three *fixed* frequency anchors were taken once, from the fine grid already computed during local refinement, bracketing that feature’s discrete extremum:

$$\begin{aligned} A_{2,0} &: [5.70, 5.80, 5.90], \\ A_{0,1} &: [5.50, 5.65, 5.80], \\ A_{1,1} &: [6.45, 6.60, 6.75], \\ A_{2,1} &: [4.90, 4.95, 5.00], \\ A_{0,0} &: [7.80, 7.85, 7.90]. \end{aligned} \quad (3)$$

These three frequencies per feature are frozen and never re-derived: for any trial (β, γ, R) , the EMM field is re-solved and re-projected onto the relevant single mode at exactly these anchor frequencies,

giving three complex samples $A_{n,j}(\omega_1)$, $A_{n,j}(\omega_2)$, $A_{n,j}(\omega_3)$. The exact quadratic through

$$y(\omega) = a\omega^2 + b\omega + c, \quad y = |A_{n,j}(\omega)|^2, \quad (4)$$

is then fitted by Newton divided differences (valid for unequally spaced anchors, which several of the five windows in Equation (3) are), and the vertex

$$\omega_\star = -\frac{b}{2a} \quad (5)$$

is taken as the continuous, sub-grid feature location. Fitting $y = |A|^2$ rather than $|A|$ is deliberate: it is smooth around both a maximum and a near-zero, and avoids the cusp that $|A|$ itself develops at a genuine zero. Because the anchors are fixed rather than re-searched, the resulting observable $\omega_\star(\beta, \gamma, R)$ varies smoothly with the parameters, which is the property a finite-difference derivative actually needs.

Validity checks, not silent fallbacks. Three diagnostic conditions are checked at every evaluation: *degenerate* (the three points are numerically too flat to fit a usable parabola, so $\omega_\star$ falls back to the discrete anchor extremum rather than an interpolated value); *extrapolated* (the fitted vertex falls outside the anchor bracket, meaning the trial parameters may have moved the true feature past what the frozen anchors can resolve); and *kindMismatch* (the parabola opens the wrong way for the feature’s declared kind, maximum or near-zero, a sign that the local shape has changed character rather than merely shifted). All three are surfaced, never silently resolved, and a strict-evaluation mode is available that raises an error rather than returning a value when any of the three is true, so that a single invalid fit inside a later finite-difference loop cannot silently contaminate a derivative.

The feature extractor and its supporting quadratic fit were tested independently before being trusted against the real forward model. The quadratic fit was verified to recover a known parabola to numerical precision at unequally spaced points and to return exactly zero curvature on flat/collinear input. The three validity flags were each exercised directly against controlled synthetic (ω, y, kind) triples, with no forward-model dependency, before the full feature extractor was run end to end against the real EMM and projection pipeline.

7 The reference feature vector after sub-grid interpolation

All five features refine cleanly at the reference point. At (β_0, γ_0, R_0) , the sub-grid feature vector is

$$f_0 = \begin{bmatrix} 5.836159 \\ 5.676641 \\ 6.647953 \\ 4.959282 \\ 7.846157 \end{bmatrix} \text{ rad/s}, \quad (6)$$

in the fixed order $[A_{2,0}, A_{0,1}, A_{1,1}, A_{2,1}, A_{0,0}]$. Every one of the five fitted vertices falls strictly inside its anchor bracket from Equation (3), and all three validity flags are false for every feature, including under the strict-evaluation mode. Table 1 compares the sub-grid locations against the discrete grid-level estimates from the reconnaissance stage.

Feature	Discrete (coarse refinement)	Sub-grid $\omega_\star$	Shift
$A_{2,0}$	5.80	5.836159	+0.036
$A_{0,1}$	5.65	5.676641	+0.027
$A_{1,1}$	6.60	6.647953	+0.048
$A_{2,1}$	4.95	4.959282	+0.009
$A_{0,0}$	7.85	7.846157	−0.004

Table 1: Discrete versus sub-grid feature locations (rad/s) at the reference point. Every shift is a small fraction of its anchor bracket’s half-width.

The shift itself is not the important result. The modest size of the shifts in Table 1 is reassuring, but it is not the point of this stage. The point is that f_0 is now a value of a function $\mathbf{f}(\beta, \gamma, R)$ that varies continuously with its arguments in a neighbourhood of the reference point, rather than a value pinned to a frequency grid. That property, not the numerical closeness to the discrete estimate, is what makes a finite-difference Jacobian meaningful in the next section.

8 The sensitivity Jacobian and its step-size validation

Central finite differences. With $\mathbf{p} = [\beta, \gamma, R]^\top$, the sensitivity Jacobian is

$$J_{ij} = \frac{\partial f_i}{\partial p_j} \approx \frac{f_i(\mathbf{p}_0 + \Delta p_j \mathbf{e}_j) - f_i(\mathbf{p}_0 - \Delta p_j \mathbf{e}_j)}{2 \Delta p_j}, \quad \Delta p_j = \epsilon p_{0j}, \quad (7)$$

using a *relative*, not absolute, perturbation size ϵ for each parameter, since β , γ , and R differ from one another by several orders of magnitude. Every one of the seven feature-vector evaluations this requires, the baseline and the six perturbed evaluations, is run under the strict-evaluation mode of Section 6: if a perturbation moves any feature outside its fixed anchor bracket, or flips its curvature character, the whole Jacobian computation is aborted rather than allowed to silently return a contaminated derivative.

The Jacobian is stable across three orders of magnitude in step size. Three relative step sizes, $\epsilon = 10^{-3}, 10^{-4}, 10^{-5}$, were each run to completion at the reference point with no strict-mode failures. Table 2 reports the column-wise relative change between successive step sizes, $\|J(\epsilon_a)_{:,j} - J(\epsilon_b)_{:,j}\| / \|J(\epsilon_a)_{:,j}\|$.

Parameter	$10^{-3} \rightarrow 10^{-4}$	$10^{-4} \rightarrow 10^{-5}$
β	1.326×10^{-6}	3.306×10^{-6}
γ	6.445×10^{-7}	3.886×10^{-5}
R	1.251×10^{-4}	1.250×10^{-6}

Table 2: Column-wise relative change in the raw Jacobian between successive finite-difference step sizes.

Reading the convergence table correctly. The R column’s change shrinks by almost exactly a factor of 100 as ϵ drops by a factor of 10 ($1.251 \times 10^{-4} \rightarrow 1.250 \times 10^{-6}$), the signature of a central difference whose error is still dominated by $O(\epsilon^2)$ truncation error. The β column changes only mildly between the two step-size pairs ($1.326 \times 10^{-6} \rightarrow 3.306 \times 10^{-6}$), consistent with a derivative that has already converged and is now showing ordinary floating-point-level jitter. The γ column behaves differently: its change increases by a factor of roughly 60 going into $\epsilon = 10^{-5}$, a stronger signature of the onset of a numerical floor at the smallest step size, though the absolute size of that change, 3.886×10^{-5} , remains extremely small ($\approx 0.0039\%$). Because γ already carries the smallest raw signal of the three parameters (Section 9), this behaviour deserved a direct check rather than being dismissed; that check is reported in Section 11.

Working step size. $\epsilon = 10^{-4}$ was adopted as the working finite-difference step: past the point where truncation error dominates for all three parameters, and short of where any column shows meaningful degradation. All Jacobian values quoted from here on are at this step size unless stated otherwise.

9 Non-dimensional scaling and the scaled Jacobian

Why the raw Jacobian cannot be interpreted directly. $\beta_0 \sim 10^{-5}$, $\gamma_0 \sim 10^{-3}$, and $R_0 \sim 10^{-1}$ differ by several orders of magnitude, so a partial derivative with respect to β is numerically large purely because of the small denominator, not necessarily because the feature vector is more sensitive to β in any meaningful sense. The raw columns of J are therefore not comparable to one another.

The dimensionless scaled Jacobian. Defining

$$D_p = \text{diag}(\beta_0, \gamma_0, R_0), \quad D_f = \text{diag}(f_0), \quad (8)$$

the scaled sensitivity matrix is

$$J_s = D_f^{-1} J D_p, \quad (J_s)_{ij} = \frac{p_j}{f_i} \frac{\partial f_i}{\partial p_j} \approx \frac{\partial \ln f_i}{\partial \ln p_j}, \quad (9)$$

whose entries are approximately the fractional change in feature i per fractional change in parameter j , a genuinely comparable quantity across columns.

The scaled Jacobian at the working step size. At $\epsilon = 10^{-4}$,

$$J_s = \begin{bmatrix} -0.0210 & 0.0003 & -0.4184 \\ -0.0578 & 0.0000 & -0.2644 \\ -0.0568 & 0.0014 & -0.2580 \\ 0.0449 & -0.0004 & -0.6927 \\ 0.0739 & -0.0070 & -0.7965 \end{bmatrix}. \quad (10)$$

The γ column is uniformly small next to the β and, especially, R columns: for example the $A_{0,0}$ row gives $\Delta f_5/f_5 \approx 0.0739 \Delta \beta/\beta - 0.0070 \Delta \gamma/\gamma - 0.7965 \Delta R/R$, so a 1% relative change in R moves that feature by roughly 0.8%, while a 1% relative change in γ moves it by only about 0.007%, more than two orders of magnitude less responsive. The same pattern, R strongest, β moderate, γ weakest, holds across all five rows of Equation (10).

10 Singular-value decomposition and local identifiability

The scaled Jacobian is locally full rank. The singular values of J_s at $\epsilon = 10^{-4}$ are

$$\sigma(J_s) = [1.1949, \quad 0.112398, \quad 0.00407693]. \quad (11)$$

All three are nonzero, so the five-feature vector carries locally independent information about all three parameters at the reference point in the noise-free linearised problem: none of β , γ , R is structurally unobservable here.

But the local inverse problem is significantly ill-conditioned. The scaled condition number is

$$\kappa(J_s) = \sigma_1/\sigma_3 \approx 293, \quad (12)$$

and the weakest right singular vector, in relative (β, γ, R) coordinates, is

$$v_3 \approx [-0.0342, \quad -0.9994, \quad 0.00254]^\top. \quad (13)$$

v_3 is almost a pure γ direction. This is a stronger and more precise statement than “ γ has a small column in J_s ”: the SVD identifies the specific, least-observable *combination* of the three physical parameters, and that combination turns out to be overwhelmingly dominated by γ alone rather than some mixture of all three.

Ill-conditioned is not the same as unidentifiable. Because $\sigma_3 \neq 0$, γ should not be described as unidentifiable from this feature set; the correct statement is that it is weakly constrained. A condition number of $\kappa \approx 293$ is, loosely, the factor by which relative error in the feature vector (from measurement noise or feature-extraction error in a real SSI pipeline) can be amplified along the weakest direction: a few percent relative error in f could plausibly correspond to a much larger relative error specifically in the recovered γ , while barely affecting the recovered β or R . That practical consequence, not the bare value of κ , is the reason this result matters.

11 Robustness of the identifiability result to finite-difference step size

The weak- γ result is not a finite-difference artefact. The scaled SVD was recomputed at all three step sizes from Section 8, addressing directly the γ -column convergence behaviour flagged there. Table 3 shows the singular values and condition number are essentially unchanged.

ϵ	σ_1	σ_2	σ_3	κ
10^{-3}	1.1950	0.112396	0.0040768	293.122
10^{-4}	1.1949	0.112398	0.00407693	293.087
10^{-5}	1.1949	0.112398	0.00407712	293.073

Table 3: Scaled singular values and condition number, all three finite-difference step sizes.

The weak direction itself is identical, not merely similar. Comparing the weakest right singular vector pairwise across the three step sizes, allowing for the arbitrary sign of a singular vector,

$$|v_3(10^{-3})^\top v_3(10^{-4})| = |v_3(10^{-4})^\top v_3(10^{-5})| = |v_3(10^{-3})^\top v_3(10^{-5})| = 1.00000000 \quad (14)$$

to eight displayed decimal places. The elevated column-wise convergence metric noted for γ in Section 8 at $\epsilon = 10^{-5}$ therefore has no detectable effect on either the singular values or the identified weak direction: whatever residual numerical floor is present in the raw γ column at the smallest step size is far too small to move the scaled, rank-reduced result. The identifiability conclusion of Section 10 stands independently of which of the three tested step sizes is used.

12 Discussion

12.1 Central finding

The five-feature vector is locally structurally informative for all three unknown parameters at the reference point, but the inverse map is poorly conditioned in a direction almost entirely dominated by the mass/draft parameter γ .

Section 6 replaced the discrete, grid-quantised feature locations of the previous report with a continuous, validated feature map $\mathbf{f}(\beta, \gamma, R)$. Section 8 constructed a finite-difference sensitivity Jacobian of that map and showed it is stable across three orders of magnitude in step size. Sections 9 and 10 showed that the appropriately non-dimensionalised Jacobian is locally full rank, $\kappa \approx 293$, with a weak singular direction that is almost pure γ ; and Section 11 confirmed that this conclusion is not an artefact of the finite-difference step size used to obtain it.

12.2 What has, and has not, been established

Newly established in this stage: the three items the previous report left explicitly open have each been directly addressed. Sub-grid feature precision is resolved by the fixed-anchor quadratic interpolator of Section 6, tested independently of the forward model and then against it. Parameter sensitivity is resolved by the finite-difference Jacobian of Section 8, validated for step-size stability. Local identifiability at the reference point is resolved by the scaled SVD of Section 10: locally

full rank, significantly ill-conditioned, weak direction overwhelmingly aligned with γ , and this last result confirmed robust to finite-difference step size in Section 11.

Still not established, and explicitly out of scope for this stage:

1. *Generality beyond the reference point.* Every result in this document is local, at the single reference parameter set of Equation (2). Whether the same qualitative picture, full rank with a γ -dominated weak direction, persists across a wider, physically plausible region of (β, γ, R) has not been tested.
2. *Validity of the fixed anchors away from the reference point.* The anchors of Equation (3) were derived once, at the reference point, and are deliberately never re-searched. This is exactly what keeps the Jacobian smooth locally, but it also means the same anchors cannot simply be reused at a substantially different (β, γ, R) : a large enough parameter change would move a true feature outside its fixed bracket (the extrapolated flag of Section 6 exists precisely to catch this), at which point a fresh reconnaissance and refinement would be needed at the new point, not merely a re-evaluation of the existing anchors.
3. *Practical recovery under noise.* The condition number $\kappa \approx 293$ is a statement about noise amplification in principle, not a demonstration of what actually happens to a nonlinear parameter recovery under realistic feature-extraction error. That is a distinct, practical question from the linear-algebra result obtained here.
4. *Connection to SSI.* As in the previous report, these remain forward-model, EMM-side observables. Whether a state-space model fitted to a simulated or measured buoy time series by SSI can actually recover features at or near these same frequencies, with the accuracy this identifiability result assumes, is untested here.

12.3 Summary and next steps

The previous stage established that mode-resolved observables contain useful candidate inverse features that a global response norm hides. This stage answered the question that raised: whether those five features actually constrain (β, γ, R) . At the reference point, the answer is yes locally, but unevenly, with a scaled Jacobian that is numerically stable, full rank, and significantly better conditioned in the β and R directions than in the γ direction.

The natural next experiment is a nonlinear twin inversion: generate a synthetic feature vector from a known, nearby (β, γ, R) , hide those parameters from a bounded nonlinear optimiser, and attempt to recover them from several different starting guesses. Two variants of this experiment are needed, not one. A noise-free version, run to tight convergence, mainly checks that the optimiser reaches the correct point at all and is a weaker test of this stage's conditioning result than it might appear: with exact data and a full-rank system, even a poorly conditioned inverse problem can in principle be solved exactly, so a clean noise-free recovery of γ would not by itself contradict $\kappa \approx 293$. What the noise-free version can still show, if tracked explicitly, is slower convergence or a flatter objective specifically along the γ direction across different starting guesses. The more direct test is a second version with a small synthetic perturbation added to the feature vector (for instance 0.1–1% relative error, representative of realistic feature-extraction uncertainty), repeated over several noise realisations: this is expected, from the SVD result above, to show β and R recovered consistently while γ shows substantially larger scatter. That comparison, not the noise-free case alone, is the practical test of what this stage's identifiability result actually predicts.

13 Next Steps

The results above establish local identifiability at a reference/test parameter set, not at the actual physical floe. The remaining project work is therefore aimed at moving from this synthetic reference point to the real floe: establishing its effective R , γ and β , and then repeating the local inverse analysis around that physical parameter regime. For the composite floe, which combines polypropylene with an epoxy/glass shell, β should be understood as an effective flexural-stiffness parameter rather than a

value read from a handbook material table. The sequence below also closes the four open items listed in Section 12: practical nonlinear recovery under noise, generality away from the reference point, the bridge to SSI, and the eventual experimental comparison against the real floe.

1. **Noise-free EMM inverse twin test (immediate next step).** A known parameter set near the current reference point, $\mathbf{p}_{\text{true}} = [\beta, \gamma, R]^\top$, is chosen and its five feature frequencies are generated directly from the EMM, $\mathbf{f}_{\text{true}} = \mathbf{f}(\mathbf{p}_{\text{true}})$. The value $\mathbf{p}_{\text{true}}$ is then hidden and recovered by solving a bounded nonlinear optimisation problem,

$$\min_{\mathbf{p}} \|\mathbf{f}(\mathbf{p}) - \mathbf{f}_{\text{true}}\|^2,$$

run from several initial guesses. The purpose is to establish that the nonlinear inverse machinery can actually recover known parameters, not merely that the local Jacobian is full rank. Given the SVD result of Section 10, convergence along γ warrants particular attention, since that is the weak direction.

2. **Controlled perturbation and feature-noise inversion.** Once the exact twin test succeeds, the five target feature frequencies are perturbed by controlled amounts and the inversion is repeated over multiple random realisations. Initial perturbations should be illustrative (for example 0.1–1%) rather than claimed as representative of measured IMU noise. The main question is whether the SVD prediction of Section 10 is borne out in practice, namely R and β recovered with relative stability while γ shows substantially larger scatter. This step converts the local conditioning result obtained here into an actual parameter-recovery study, and is the more direct test of the $\kappa \approx 293$ result than the noise-free case alone.
3. **Broadening the EMM inverse analysis beyond one reference floe.** The present result is deliberately local. A physically plausible region of (β, γ, R) is sampled to test whether the same modal features persist, whether the Jacobian remains full rank, and whether γ remains the weak direction throughout that region. The fixed anchor windows of Equation (3) will not necessarily remain valid after large parameter changes, so this stage will likely require adaptive feature tracking or local re-reconnaissance rather than reusing the current anchors everywhere.
4. **Characterising the actual IMU noise.** The buoy IMUs are recorded while stationary and their noise is quantified: bias, variance and RMS noise, power spectral density versus frequency, low-frequency drift, axis dependence, and, where relevant, sensor-to-sensor differences. This produces an empirical noise model in place of an arbitrary Gaussian assumption, to be injected into synthetic sensor signals ahead of SSI validation. This step is distinct from Item 2: Item 2 tests the inverse feature-to-parameter map, whereas this step characterises the measurement system itself.
5. **Completing SSI validation under realistic sensor noise and imperfections.** Synthetic multi-sensor time series from the forward model are contaminated using the measured IMU noise model of Item 4, and the existing SSI machinery, model-order sweeps, stability classification, branch persistence, and MAC comparisons, is rerun on the contaminated data. The question changes from whether SSI works under ideal synthetic data to which modes and features remain recoverable at realistic signal-to-noise ratio. Earlier sensor-placement work becomes directly relevant here, since the actual four-sensor arrangement and the noise model together determine what is observable.
6. **JONSWAP and broadband forcing study.** The deliberately simple single-tone and two-tone forcing used so far is replaced with a realistic irregular-wave excitation drawn from a JONSWAP spectrum, fed through the forward model to produce multi-sensor floe responses, to which output-only SSI is then applied. The questions of interest are whether SSI recovers persistent modal content under broadband stochastic excitation, how accurately the modal frequencies and shapes are recovered, and how that accuracy depends on spectrum, record length, sensor placement and IMU noise. This is a substantially more realistic test than the pure-tone validation used previously.
7. **Establishing the SSI-to-inversion bridge.** This is one of the most important remaining items, and is the fourth open point of Section 12. The inverse features used in this document are forced

EMM modal-response maxima and near-zeros, while SSI identifies state-space poles and modes from time-series data; the two are not automatically the same quantities. This step determines which SSI-recoverable observables can actually be used for inversion. Several outcomes are possible: some EMM feature frequencies may correspond closely to SSI-recovered modal frequencies, some may not, and near-zero forced-response features may be particularly difficult or impossible to obtain from output-only SSI. The final inverse feature vector may therefore differ from the current five-feature EMM vector.

8. **SSI-driven parameter inversion.** Once the bridge in Item 7 is established, the exact EMM feature vector used in the inverse problem so far is replaced with features extracted from SSI,

$$\text{sensor time histories} \longrightarrow \text{SSI} \longrightarrow \hat{\mathbf{f}} \longrightarrow (\widehat{\beta, \gamma, R}).$$

This is first carried out with fully synthetic data, for which the true parameters are known, and then repeated with measured IMU-noise corruption and JONSWAP forcing. This constitutes the end-to-end synthetic twin test of the full pipeline.

9. **Characterising the physical floe.** Independent ground-truth estimates of the physical parameters are obtained. R follows from the SolidWorks geometry or a measured planform. γ is established from total mass and planform area rather than by treating the composite floe as a homogeneous plate,

$$\gamma = \frac{m/A}{\rho_w H}.$$

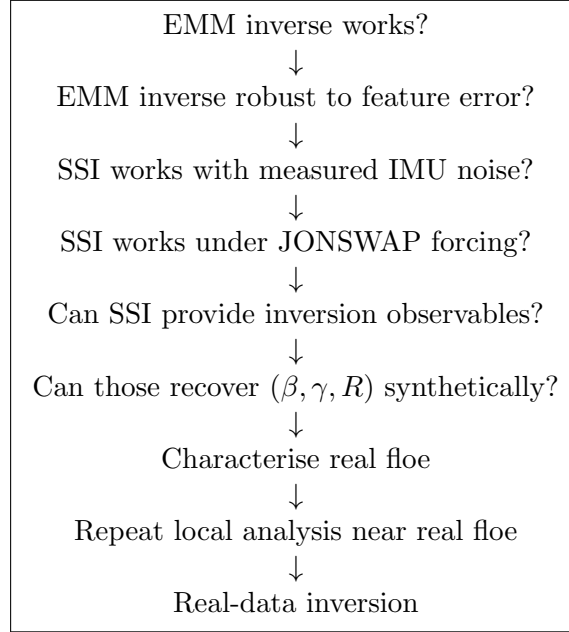
β is harder to obtain directly,

$$\beta = \frac{D_{\text{eff}}}{\rho_w g H^4},$$

where D_{eff} is the effective bending rigidity of the actual composite floe. Because the floe is polypropylene with an epoxy/glass shell, a structural bending or modal test is likely to be more reliable for D_{eff} than handbook values of E and ν .

10. **Repeating the reconnaissance and Jacobian analysis around the real floe.** Once approximate physical values of R , γ and β are available from Item 9, the EMM is run around that point without assuming that the current $A_{n,j}$ curves or anchor frequencies of Equation (3) remain applicable. The modal sweep is repeated, suitable features and anchors are identified in the physical regime, the scaled Jacobian $\mathbf{J}_s$ is recomputed, and the SVD is redone. This determines whether the actual floe lies in a well- or poorly-identifiable part of parameter space.
11. **Final experimental validation.** The IMUs are deployed on the actual floe, real response data is collected, and the same signal-processing and SSI pipeline is run to recover physical parameters. Those inverse estimates are then compared against the independently measured R , γ and β_{eff} of Item 9. This is the experimental closing of the loop for the project.

The overall remaining project can be read as the following sequence of gating questions:



The immediate next step remains Item 1, the noise-free EMM twin inversion. SSI and JONSWAP-forcing work should not be attempted before that step is complete: the sequence is chosen deliberately so that the nonlinear inverse machinery is validated in isolation first, with feature-level noise, measured IMU noise, broadband forcing, SSI, and finally the real floe added one at a time rather than all at once.